

Assignment 1: AI System Specification

Course: Introduction to Artificial Intelligence

Topic: An Automated Warehouse Sorting Robot (e.g., Amazon Robotics)

Task 1: PEAS Specification

The Automated Warehouse Sorting Robot acts as an autonomous agent tasked with navigating a complex grid-based warehouse to retrieve items efficiently.

Performance Measures:

Path Cost: Total distance traveled from charging dock to shelf.

Nodes Expanded: Total grid squares explored during search.

Runtime: Time taken to compute the optimal path.

Safety Compliance: Successful navigation without colliding with shelving units (obstacles).

Environment: A 2D grid-based warehouse layout containing static shelving units, narrow aisles, and designated start/end points.

Actuators: Motor control system for grid-based movement (Up, Down, Left, Right).

Sensors: Grid state matrix identifying empty tiles, obstacles, and pathfinding coordinates.

Task 2: Environment Classification

Observability: Fully Observable

Justification: The agent has a complete grid map of the warehouse layout at all times.

Agents: Single-agent

Justification: The robot operates alone in its assigned grid space to retrieve specific shelf items.

Determinism: Deterministic

Justification: Given a specific grid layout, an algorithm will always produce the same path and cost.

Episodicity: Episodic

Justification: Each retrieval mission is independent; the robot's success in one mission does not inherently change the warehouse structure for the next.

Dynamism: Static

Justification: Shelving units and obstacles do not move while the agent is computing or executing its path.

Continuity: Discrete

Justification: The agent moves between distinct, predefined grid cells.

Task 3: Critical Analysis

1. Greatest Technical Challenge

The Path Cost vs. Runtime trade-off is the greatest challenge. While BFS guarantees the shortest path in terms of steps, algorithms like A* or UCS must balance the computational cost (nodes expanded) against the quality of the path. Implementing efficient heuristics for A* to navigate narrow aisles without excessive exploration is technically demanding.

2. Utility Function and Trade-off

Proposed Utility Function: $U = -(\text{Path Cost} + \lambda \times \text{Runtime})$.

Trade-off Analysis: This function penalizes both distance (Path Cost) and computing time (Runtime), with λ as a weighting factor. If the weight of Runtime were doubled, the agent would prioritize faster, "greedy" paths—potentially sacrificing the shortest possible route to reduce calculation time, which might lead to inefficient long-term energy consumption.

Task 4: Structured Data Design (JSON)

```
{
  "system_name": "Warehouse Robot Pathfinding",
  "peas": {
    "performance_measures": ["Path_Cost", "Nodes_Expanded", "Runtime", "Safety"],
    "environment": "Grid-based warehouse layout",
    "actuators": ["Movement_motors"],
    "sensors": ["Grid_map_data"]
  },
  "utility_function": "U = -(Path_Cost + λ * Runtime)"
}
```